iDABET 產學研工程人才實務能力卓越基地計畫 DEVELOPMENT APPLICATION BASE OF ENGINEERING TALENTS

The Development Application Base of Engineering Talents (iDABET)

Project of the Industrial Development Administration, Ministry of Economic Affairs

2025 Practical Ability Development Implementation Unit Project – Engineering Talent Practical Ability Development Record Form

Project No. of the Practical Ability Development Implementation Unit: DABET11403

No. of Engineering Talent: 40

Name of the Practical Ability Development Implementation Unit: 財團法人金屬工業研究發展中心/光電技術組

Project Name of the Practical Ability Development Implementation Unit: 半導體產業之真空設備與製程能力發展與人工智慧控制應用

Name of the Engineering Talent: 蘇翼

Item	**Date**	**Instructor**	**No. of Hours** (Note 1)	**Type of Practical Ability Development Topic** (Note 2)	**Topic**	**Description**
1	3/4	王聖禾	4	1	Indoor Positioning System Research(Thesis)	Analyzing past papers and journals that explore the use of indoor positioning technology.
2	3/7	王聖禾 / 李宗信	3	1	Project Briefing	Discussion of the project topic and briefing about the goals and expectations of the internship.
3	3/8	王聖禾	6	1	Indoor Positioning System Research(Thesis)	Optimizing the design of the hexapod robot focusing on mechanical structure, locomotion algorithms, power efficiency and sensor integration to use in indoor positioning.
4	3/13	李宗信	2	2	Introduction To Lithography	Lithography is used in semiconductor manufacturing to create intricate patterns on a substrate.
5	3/13	黃乘敦	2	1	Clean room classifications / Environment and safety management	Introducing the classification of the clean rooms (ISO 1-9, 9 being the least clean). Handling of waste gas from the laboratory.
6	3/14	王聖禾	3	1	Project Briefing	Implementation of a hexapod robot within indoor positioning.

7	3/17	郭維甫	3	1	Indoor Positioning System Research(Thesis)	Analyzing the movement of the robotic arm tool, the tool reference point system can also serve as a reference for the walking dynamics of the hexapod.
8	3/18	王聖禾	6	1	Indoor Positioning System Research(Thesis)	Analyzing past papers and journals that uses inverse kinematic to operate a 3DoF robot arm.
9	3/20	張益三	2	2	Fluid mechanics in the Semiconductor industry	Introduction to Fluid mechanics and how it plays a crucial role in determining the thickness, uniformity, and quality of the coated film.
10	3/20	李宗信	2	1	Applications of PECVD, CVD & PVD / Applications of microwaves	Applications of coating photoresist on a substrate. The concept and use of a microwave.
11	3/21	王聖禾	3	1	Project Briefing	Stabilizing the hexapod robot to maintain a level body on any terrain.
12	3/22	王聖禾	6	1	Indoor Positioning System Research(Thesis)	Analyzing a past senior's research in indoor positioning.
Total Hours on Practical Ability Development Topics			Total ____42_____ hours.			

Please add rows to the table as needed.

Notes: (1) The hours for implementing practical ability development topics should be filled in according to the actual implementation status between March 1 and August 31.

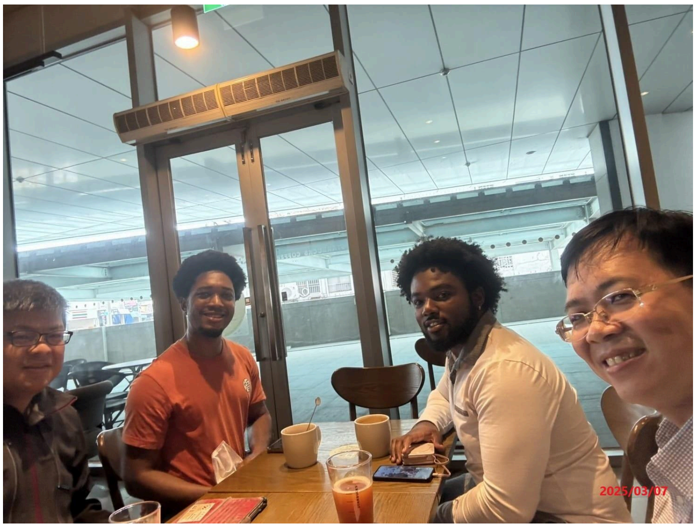
(2) Type of practical ability development topics: 1. Practice-based; 2. Industry collaboration-based.

(3) Please attach at least 6 dated photos (yyyy/mm/dd) as evidence of the engineering talent's participation in practical ability development topics or activities.

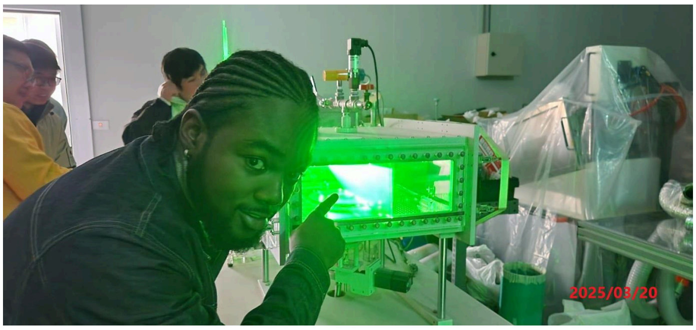
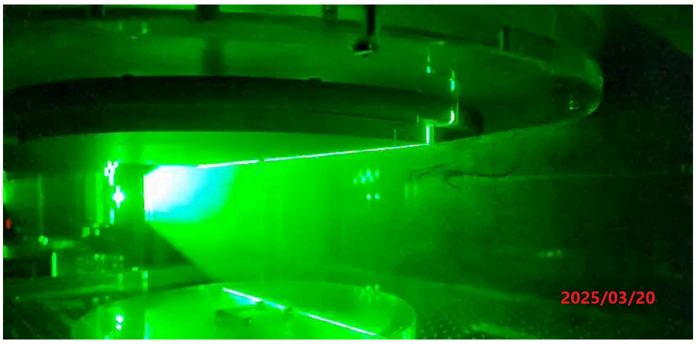
(4) Both the instructor and engineering talent shall sign at the bottom of the record form.

(5) To implement the “2025 Development Application Base of Engineering Talents (iDABET) Project”, the Industrial Development Administration, Ministry of Economic Affairs will collect, process and use the personal data of the party involved without notifying the said party, as prescribed in Subparagraph 2 of Paragraph 2 of Article 8 of the Personal Data Protection Act.

【Photos of Participation in Practical Ability Development Topics or Activities】

 Project and Internship Briefing	The journey of adapting to the workforce in Taiwan as a foreigner begins here. We reviewed the project topics and outlined the objectives for the upcoming months.
 Environment and safety Management: Air filtration system	Today, we explored the filtration process of the scrubber, which operates through multiple stages to ensure the release of cleaner air into the environment. These systems are highly effective at capturing fine dust particles, removing over 99% of airborne particulate matter by trapping them in liquid droplets.

 Gas Cabinet Supply System	This is a gas cabinet. It is a ventilated, safety-enclosed system designed to store and control the delivery of hazardous, toxic, or flammable gases in industrial and laboratory settings. It ensures safe handling, and distribution of gases while minimizing risks such as leaks, fire, or contamination.
 Introduction to Spin coating Photoresist	Spin coating is a method used in many manufacturing processes such as semiconductor fabrication. It applies a thin, uniform layer of photoresist onto a substrate, usually a silicon wafer. This process uses centrifugal forces to spread the liquid resist evenly across the surface.

  Application of Fluid mechanics in Spin Coating Photoresist	This machine features a specialized shower head for the spin coating process, which affects the gas flow within the chamber. A laser beam slices through the volume, creating a defined area for fluid mechanics analysis. This approach enables precise control over film uniformity, minimizes defects, and ensures high-resolution patterning.
 Introduction and Application of Microwaves	Here, we have a standard microwave, which heats substances by agitating water molecules through electromagnetic waves (produced by the magnetron in my hand) typically at 2.45 GHz. Unlike convection heating, where heat is transferred from the surrounding environment to the target, microwaves directly penetrate the substance, causing molecular vibrations that generate heat internally, resulting in a more even and efficient heating process.

Notes:

1. Please add rows to the table as needed. Each photo shall be dated at the bottom right, and each photo shall be accompanied by descriptions below it to explain the topic.
2. Please provide the original files of 1 to 3 photos (JPEG files, each above 1MB in size and with a resolution above 300 DPI).